Smart Campus Management System

Java Console-Based Project | GitHub Portfolio

GitHub Repository Name

Smart-Campus-Management-System

Project Overview

The Smart Campus Management System is a Java application that helps manage basic student information in a college. It allows users to add student records, view all records, search for a student using an ID, and check attendance eligibility through a simple menu-driven interface.

Main Features

- Add student details such as ID, name, department, and attendance.
- Display all stored student records.
- Search for a student using the student ID.
- Check whether attendance is at least 75%.
- Use a simple menu that repeats until the user chooses Exit.

Technologies and Concepts

Language: Java
Concepts: Classes, Objects, Constructors, ArrayList, Methods, Loops, Switch Statement, If-Else, Scanner, and Encapsulation basics.

How to Run

```
1. Save the file as SmartCampusManagementSystem.java
2. Open a terminal in that folder.
3. Compile: javac SmartCampusManagementSystem.java
4. Run:     java SmartCampusManagementSystem
```

Complete Java Source Code

```
import java.util.ArrayList;
import java.util.Scanner;

class Student {
    int id;
    String name;
    String department;
    double attendance;

    Student(int id, String name, String department, double attendance) {
        this.id = id;
        this.name = name;
        this.department = department;
        this.attendance = attendance;
    }

    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Department: " + department);
```

```
        System.out.println("Attendance: " + attendance + "%");
        System.out.println("----------------------------");
    }
}

public class SmartCampusManagementSystem {
    static ArrayList<Student> students = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);

    static void addStudent() {
        System.out.print("Enter student ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter student name: ");
        String name = sc.nextLine();

        System.out.print("Enter department: ");
        String department = sc.nextLine();

        System.out.print("Enter attendance percentage: ");
        double attendance = sc.nextDouble();

        students.add(new Student(id, name, department, attendance));
        System.out.println("Student added successfully!");
    }

    static void viewStudents() {
        if (students.isEmpty()) {
            System.out.println("No student records found.");
            return;
        }

        System.out.println("\n--- STUDENT RECORDS ---");
        for (Student student : students) {
            student.display();
        }
    }

    static void searchStudent() {
        System.out.print("Enter student ID to search: ");
        int searchId = sc.nextInt();

        for (Student student : students) {
            if (student.id == searchId) {
                System.out.println("Student found:");
                student.display();
                return;
            }
        }

        System.out.println("Student not found.");
    }

    static void showAttendanceStatus() {
        System.out.print("Enter student ID: ");
        int searchId = sc.nextInt();

        for (Student student : students) {
            if (student.id == searchId) {
                System.out.println("Attendance: " + student.attendance + "%");

                if (student.attendance >= 75) {
                    System.out.println("Status: Eligible");
                } else {
                    System.out.println("Status: Attendance shortage");
                }
                return;
            }
        }

        System.out.println("Student not found.");
    }

    public static void main(String[] args) {
        int choice;

        do {
            System.out.println("\n===== SMART CAMPUS MANAGEMENT SYSTEM =====");
            System.out.println("1. Add Student");
```

```
            System.out.println("2. View All Students");
            System.out.println("3. Search Student");
            System.out.println("4. Check Attendance Status");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");

            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    addStudent();
                    break;
                case 2:
                    viewStudents();
                    break;
                case 3:
                    searchStudent();
                    break;
                case 4:
                    showAttendanceStatus();
                    break;
                case 5:
                    System.out.println("Thank you for using the system!");
                    break;
                default:
                    System.out.println("Invalid choice. Try again.");
            }
        } while (choice != 5);

        sc.close();
    }
}
```

Note: This is a beginner-friendly console project. Student records are stored in memory and will be cleared when the program closes. It does not use a database or a graphical interface.